Hence the limit does not exist. Note that $f(0^-) \neq f(0^+)$.

(ii) $$\frac{\sin x}{|x|} = \begin{cases} \dfrac{\sin(x)}{-x} & \text{if } -1 < x < 0 \\ \dfrac{\sin x}{x} & \text{if } 0 < x < 1 \end{cases}$$

Therefore, $\lim\limits_{x \to 0^-} f(x) = -1$

$\lim\limits_{x \to 0^+} f(x) = 1$

Hence the limit does not exist.

(iii) $$f(x) = \frac{x\lfloor x \rfloor}{\sin|x|} = \begin{cases} \dfrac{-x}{\sin(-x)} & \text{if } -1 < x < 0 \\ \dfrac{x.0}{\sin x} & \text{if } 0 < x < 1 \end{cases}$$

$$= \begin{cases} \dfrac{x}{\sin x} & \text{if } -1 < x < 0 \\ 0 & \text{if } 0 < x < 1 \end{cases}$$

Therefore, $\lim\limits_{x \to 0^-} f(x) = +1$

$\lim\limits_{x \to 0^+} f(x) = 0.$

Hence the limit does not exist.

(iv) $$\frac{\sin\left(x - \lfloor x \rfloor\right)}{x - \lfloor x \rfloor} = \begin{cases} \dfrac{\sin(x-(-1))}{x-(-1)} & \text{if } -1 < x < 0 \\ \dfrac{\sin(x-0)}{x-0} & \text{if } 0 < x < 1 \end{cases}$$

$$f(x) = \begin{cases} \dfrac{\sin(x+1)}{(x+1)} & \text{if } -1 < x < 0 \\ \dfrac{\sin x}{x} & \text{if } 0 < x < 1 \end{cases}$$

$\lim\limits_{x \to 0^-} f(x) = \dfrac{\sin 1}{1} = \sin 1$

$\lim\limits_{x \to 0^+} f(x) = 1.$

Hence the limit does not exist.

EXERCISE 9.4

Evaluate the following limits:

(1) $\lim\limits_{x \to \infty} \left(1 + \dfrac{1}{x}\right)^{7x}$

(2) $\lim\limits_{x \to 0} (1+x)^{1/3x}$

(3) $\lim\limits_{x \to \infty} \left(1 + \dfrac{k}{x}\right)^{\frac{m}{x}}$

(4) $\lim\limits_{x \to \infty} \left(\dfrac{2x^2+3}{2x^2+5}\right)^{8x^2+3}$

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